



Review

47%**29**

Out of 62 points

2:59

Avg Time Per Question

Your Answers:

Vignette

Adrienne Yu is an analyst with an international bank. She analyzes Ambieu S.A. ("Ambieu"), a multinational corporation, for a client presentation. Ambieu complies with IFRS, and its presentation currency is the Norvoltian krone (NVK). Ambieu's two subsidiaries, Ngcorp and Cendaró, have different functional currencies: Ngcorp uses the Bindiar franc (FB) and Cendaró uses the Crenland guinea (CRG).

Yu first analyzes the following three transactions to assess foreign currency transaction exposure:

Transaction 1:

Cendaró sells goods to a non-domestic customer that pays in dollars on the purchase date.

Transaction 2:

Ngcorp obtains a loan in Bindiar francs on 1 June 2016 from a European bank with the Norvoltian krone as its presentation currency.

Transaction 3:

Ambieu imports inventory from Bindiar under 45-day credit terms, and the payment is to be denominated in Bindiar francs.

Yu then reviews Transactions 2 and 3. She determines the method that Ambieu would use to translate Transaction 2 into its 31 December 2016 consolidated financial statements. While analyzing Transaction 3, Yu notes that Ambieu purchased inventory on 1 June 2016 for FB27,000/ton. Ambieu pays for the inventory on 15 July 2016. Exhibit 1 presents selected economic data for Bindiar and Crenland.

Exhibit 1

Selected Economic Data for Bindiar and Crenland

Date	Spot	Bindiar Inflation Rate (%)	Spot CRG/NVK Exchange Rate	Crenland	
	FB/NVK Exchange Rate			Inflation Rate (%)	Crenland GPI
31 Dec 2015	—	—	5.6780	—	100.0
1 Jun 2016	4.1779	—	—	—	—
15 Jul 2016	4.1790	—	—	—	—
31 Dec 2016	4.2374	3.1	8.6702	40.6	140.6
Average 2016	4.3450	—	—	—	—
31 Dec 2017	4.3729	2.1	14.4810	62.3	228.2
Average 2017	4.3618	—	11.5823	—	186.2

Prior to reviewing the 2016 and 2017 consolidated financial statements of Ambieu, Yu meets with her supervisor, who asks Yu the following two questions:

Question 1:

Would a foreign currency translation loss reduce Ambieu's net sales growth?

Question 2:

According to IFRS, what disclosures should be included relating to Ambieu's treatment of foreign currency translation for Ngcorp?

To complete her assignment, Yu analyzes selected information and notes from Ambieu's 2016 and 2017 consolidated financial statements, presented in Exhibit 2.

Exhibit 2

Selected Information and Notes from Consolidated Financial Statements of Ambieu S.A. (in NVK millions)

Income Statement	2017	2016	Balance Sheet	2017	2016
Revenue ⁽¹⁾	1,069	1,034	Cash ⁽³⁾	467	425
Profit before tax	294	269	Intangibles ⁽⁴⁾	575	570
Income tax expense ⁽²⁾	-96	-94	—	—	—

Income Statement 2017 2016 Balance Sheet 2017 2016

Net profit	198	175	—	—	—
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Note 1: Cendaro's revenue for 2017 is CRG125.23 million.

Note 2: Ambleu's consolidated income tax rate decreases by 2.29%, from 34.94% (=94/269) in 2016 to 32.65% (=96/294) in 2017. The largest component of the decrease stems from the 1.42% change in the effect of tax rates in non-domestic jurisdictions, which lowers Ambleu's consolidated income tax rate in 2016 by 3.34% (=9/269) and in 2017 by 4.76% (=14/294).

	2017 (in NVK millions)	2016 (in NVK millions)
Reconciliation of Income Tax Expense		
Income tax at Ambleu's domestic tax rate	102	92
Effect of tax rates on non-domestic jurisdictions	-14	-9
Unrecognized current year tax losses	8	11
Income tax expense	96	94

Expand / Collapse Extended Description

A table for comparison of reconciliation of income tax expense for Ambleu in years 2017 and 2016 expressed in NVK millions. Income tax at Ambleu's domestic tax rate. 2017, 102. 2016, 92. Effect of tax rates on non-domestic jurisdictions. 2017, negative 14. 2016, negative 9. Unrecognized current year tax losses. 2017, 8. 2016, 11. Income tax expense. 2017, 96. 2016, 94.

Note 3: The parent company transferred NVK15 million to Cendaro on 1 January 2016 to purchase a patent from a competitor for CRG85.17 million.

Note 4: The 2016 consolidated balance sheet includes Ngcorp's total intangible assets of NVK3 million, which were added to Ngcorp's balance sheet on 15 July 2016.

1 Multiple Choice 0 / 1 point

Which transaction would generate foreign currency transaction exposure for Ambleu?



☒ A. Transaction 1

Correct Answer:

C. Transaction 3

☐ B. Transaction 2

☐ C. Transaction 3

Feedback

C is correct. In Transaction 3, the payment for the inventory is due in Bindiar francs, a different currency from the Norvoltian krone, which is Ambleu's presentation currency. Because the import purchase (account payable) is under 45-day credit terms, Ambleu has foreign currency transaction exposure. The payment is subject to fluctuations in the FB/NVK exchange rate during the 45-day period between the sale and payment dates. Thus, Ambleu is exposed to potential foreign currency gains if the Bindiar franc weakens against the Norvoltian krone or foreign currency losses if the Bindiar franc strengthens against the Norvoltian krone.

2 Multiple Choice 0 / 1 point

Yu's determination regarding Transaction 2 should be based on the currency of the:

☐ A. loan.



B.
bank.

Correct
Answer:

C. borrower.



C. borrower.

Feedback

C is correct. The currency of Ngcorp as the borrowing foreign subsidiary, relative to that of Ambleu, determines Ambleu's choice of translation method for Transaction 2. Because Ngcorp's functional currency is the Bindian franc and Ambleu's presentation currency is the Norvoltian krone, the current rate method rather than the temporal method should be used. Regardless of the currency in which the loan is denominated, the loan is first recorded in Ngcorp's financial statements. Then, Ngcorp's financial statements, which include the bank loan, are translated into Ambleu's consolidated financial statements.



3 Multiple Choice 0 / 1 point

Based on Exhibit 1, what is the foreign exchange gain resulting from Transaction 3 on the 31 December 2016 financial statements?



A. NVK1.70 per ton



B. NVK90.75 per
ton

Correct
Answer:

A. NVK1.70 per ton



C. NVK248.54 per ton

Feedback

A is correct. On Ambleu's balance sheet, the cost included in the inventory account is the translation of FB27,000/ton into Norvoltian krone on the purchase date. Ambleu could have paid this amount on the purchase date but chose to wait 45 days to settle the account. The inventory cost is determined using the FB/NVK exchange rate of 4.1779 on the purchase date of 1 June 2016. $\text{FB}27,000/\text{FB}4.1779/\text{NVK} = \text{NVK}6,462.58/\text{ton}$

The cash outflow is the amount exchanged from the Norvoltian krone to the Bindian franc to pay the FB27,000/ton owed for the inventory 45 days after the transaction date. This payment uses the FB/NVK exchange rate of 4.1790 on the settlement date of 15 July 2016.

$$\text{FB } 27,000/\text{FB}4.1790 \text{ per NVK} = \text{NVK}6,460.88/\text{ton}$$

$$\begin{aligned} \text{Foreign exchange gain} &= \text{Inventory cost} - \text{Cash payment} \\ &= \text{NVK}6,462.58 - \text{NVK}6,460.88 \\ &= \text{NVK}1.70/\text{ton} \end{aligned}$$

Thus, Ambleu's cash outflow is less than the cost included in the inventory account, and NVK1.70/ton is the realized foreign exchange gain relating to this transaction. By deferring payment for 45 days, and because the Bindian franc decreased in value during this period, Ambleu pays NVK1.70/ton less than the inventory cost on the purchase date of 1 June 2016. Thus, Ambleu will report a foreign exchange gain in its 2016 net income.

4 Multiple Choice 1 / 1 point

What is the *best* response to Question 1?

☒ A.
Yes

☐ B. No, because it would reduce organic sales growth

☐ C. No, because it would reduce net price realization and mix

Feedback

A is correct. Net sales growth equals organic sales growth plus or minus the effects of acquisitions, divestitures, and foreign exchange. A foreign currency translation loss would reduce net sales growth. Thus the answer to Question 1 is yes.

5 Multiple Choice 1 / 1 point

Based on Exhibit 1, the *best* response to Question 2 is that Ambleu should disclose:

☐ A. a restatement for local inflation.

☐ B. that assets carried at historical cost are translated at historical rates.

☒ C. the amount of foreign exchange differences included in net income.

Feedback

C is correct. IFRS requires that Ambleu disclose “the amount of exchange differences recognized in profit or loss” when determining net income for the period. Because companies may present foreign currency transaction gains and losses in various places on the income statement, it is useful for companies to disclose both the amount of transaction gain or loss that is included in income as well as the presentation alternative used.

6 Multiple Choice 1 / 1 point

Based on Exhibit 1 and Note 1 in Exhibit 2, the amount that Ambleu should include in its 31 December 2017 revenue from Cendaró is *closest* to:

☒ A. NVK10.60 million.

☐ B. NVK13.25 million.

☐ C. NVK19.73 million.

Feedback

A is correct. Crenland experienced hyperinflation from 31 December 2015 to 31 December 2017, as shown by the General Price Index, with cumulative inflation of 128.2% during this period. According to IFRS, Cendaró’s financial statements must be restated for local inflation, then

translated into Norvoltian kroner using the current exchange rate. The 2017 revenue from Cendaró that should be included in Ambieu's income statement is calculated as follows:

Revenue in CRG $\times$ (GPI 31 December 2017/GPI average 2017) = Inflation-adjusted revenue in CRG

CRG125.23 million $\times$ (228.2/186.2) = CRG153.48 million

Inflation-adjusted revenue in CRG/31 December 2017 exchange rate (CRG/NVK) = Revenue in Norvoltian kroner

CRG153.48 million/14.4810 = NVK10.60 million

7 Multiple Choice 0 / 1 point

Based on Exhibit 2 and Note 2, the change in Ambieu's consolidated income tax rate from 2016 to 2017 *most likely* resulted from a:

- ☐ A. decrease in Ambieu's domestic tax rate.
- ☐ B. more profitable business mix in its subsidiaries.
- ☒ C. stronger Norvoltian krone relative to the currencies of its subsidiaries.

Correct Answer:

B. more profitable business mix in its subsidiaries.

Feedback

B is correct. The consolidated income tax rate is calculated as income tax expense divided by profit before tax. Note 2 shows that Ambieu's consolidated income tax rate decreases by 2.29%, from 34.94% (=94/269) in 2016 to 32.65% (=96/294) in 2017. The largest component of the decrease stems from the 1.42% change in the effect of tax rates in non-domestic jurisdictions, which lowers Ambieu's consolidated income tax rate in 2016 by 3.34% (=9/269) and in 2017 by 4.76% (=14/294). The decrease in 2017 could indicate that Ambieu's business mix shifted to countries with lower marginal tax rates, resulting in a lower consolidated income tax rate and more profit. (The change could also indicate that the marginal tax rates decreased in the countries in which Ambieu earns profits.)

8 Multiple Choice 0 / 1 point

Based on Exhibit 1 and Note 3 in Exhibit 2, the cumulative translation loss recognized by Ambieu related to the patent purchase on the 31 December 2017 financial statements is *closest* to:

- ☒ A. NVK0.39 million.

Correct Answer:

B. NVK1.58 million

- ☐ B. NVK1.58 million
- ☐ C. NVK9.12 million.

Feedback

B is correct. IAS 29 indicates that a cumulative inflation rate approaching or exceeding 100% over three years would be an indicator of hyperinflation. Because the cumulative inflation rate for 2016 and 2017 in Crenland was 128.2%, Cendaró's accounts must first be restated for local inflation. Then, the inflation-restated Crenland guinea financial statements can be translated into Ambleu's presentation currency, the Norvoltian krone, using the current exchange rate.

Using this approach, the cumulative translation loss on 31 December 2017 for the CRG85.17 million patent purchase is –NVK1.58 million, as shown in the following table.

Date	Inflation Rate (%)	Restated Carrying Value (CRG/MM)	Current Exchange Rate (CRG/NVK)	Translated Amount (NVK MM)	Annual Translation Gain/Loss (NVK MM)	Cumulative Translation Gain/Loss (NVK MM)
1 Jan 2016	—	85.17	5.6780	15.00	N/A	N/A
31 Dec 2016	40.6	119.75	8.6702	13.81	–1.19	–1.19
31 Dec 2017	62.3	194.35	14.4810	13.42	–0.39	–1.58

9 Multiple Choice 0 / 1 point

Based on Exhibit 1 and Note 4 in Exhibit 2, the total intangible assets on Ngcorp's balance sheet as of 31 December 2016 are *closest* to:

☒ A. FB12.54 million.

Correct Answer: **B. FB12.71 million.**

☐ B. FB12.71 million.

☐ C. FB13.04 million.

Feedback

B is correct. Because Ngcorp has a functional currency that is different from Ambleu's presentation currency, the intangible assets are translated into Norvoltian kroner using the current rate method. The current FB/NVK exchange rate is 4.2374 as of 31 December 2016. Thus, the intangible assets on Ngcorp's 2016 balance sheet are NVK3 million $\times$ FB4.2374/NVK = FB12.71 million.

Vignette

Triofind, Inc. (Triofind), based in the country of Norvolt, provides wireless services to various countries, including Norvolt, Borliand, Abuelio, and Certait. The company's presentation currency is the Norvolt euro

(NER), and Triofind complies with IFRS. Triofind has two wholly owned subsidiaries, located in Borliand and Abuelio. The Borliand subsidiary (Trioind-B) was established on 30 June 2016, by Trioind both investing NER1,000,000, which was converted into Borliand dollars (BRD), and borrowing an additional BRD500,000.

Marie Janssen, a financial analyst in Trioind's Norvolt headquarters office, translates Trioind-B's financial statements using the temporal method. Non-monetary assets are measured at cost under the lower of cost or market rule. Spot BRD/NER exchange rates are presented in Exhibit 1, and the balance sheet for Trioind-B is presented in Exhibit 2.

Exhibit 1

Spot BRD/NER Exchange Rates

Date	BRD per NER
30 June 2016	1.15
Weighted-average rate when inventory was acquired (2016)	1.19
31 December 2016	1.20
Weighted-average rate when inventory was acquired (2017)	1.18
30 June 2017	1.17

Exhibit 2

Trioind-B Balance Sheet for 2016 and 2017 (BRD)

	31 December 2016	30 June 2017	Liabilities and Stockholders' Equity	31 December 2016	30 June 2017
Assets					
Cash	900,000	1,350,000	Notes payable	500,000	500,000
Inventory	750,000	500,000	Common stock	1,150,000	1,150,000
			Retained earnings		200,000
Total	1,650,000	1,850,000	Total	1,650,000	1,850,000

Janssen next analyzes Trioind's Abuelio subsidiary (Trioind-A), which uses the current rate method to translate its results into Norvolt euros. Trioind-A, which prices its goods in Abuelio pesos (ABP), sells mobile phones to a customer in Certait on 31 May 2017 and receives payment of 1 million Certait rand (CRD) on 31 July 2017.

On 31 May 2017, Triofind-A also received NER50,000 from Triofind and used the funds to purchase a new warehouse in Abuelio. Janssen translates the financial statements of Triofind-A as of 31 July 2017 and must determine the appropriate value for the warehouse in Triofind's presentation currency. She observes that the cumulative Abuelio inflation rate exceeded 100% from 2015 to 2017. Spot exchange rates and inflation data are presented in Exhibit 3.

Exhibit 3

Spot Exchange Rates and Inflation Data for Triofind-A

Date	NER per CRD	NER per ABP	Abuelio Monthly Inflation Rate (%)
31 May 2017	0.2667	0.0496	—
30 June 2017	0.2703	0.0388	25
31 July 2017	0.2632	0.0312	22

Janssen gathers corporate tax rate data and company disclosure information to include in Triofind's annual report. She determines that the corporate tax rates for Abuelio, Norvolt, and Borliand are 35%, 34%, and 0%, respectively, and that Norvolt exempts the non-domestic income of multinationals from taxation. Triofind-B constitutes 25% of Triofind's net income, and Triofind-A constitutes 15%. Janssen also gathers data on components of net sales growth in different countries, presented in Exhibit 4.

Exhibit 4

Components of Net Sales Growth (%) Fiscal Year 2017

Country	Contribution from Volume Growth	Contribution from Price Growth	Foreign Currency Exchange	Net Sales Growth
Abuelio	7	6	−2	11
Borliand	4	5	4	13
Norvolt	7	3	—	10

10 Multiple Choice 0 / 1 point

Based on Exhibits 1 and 2 and Janssen's translation method, total assets for Triofind-B translated into Triofind's presentation currency as of 31 December 2016 are *closest* to:



A.
NER1,375,000.

Correct
Answer:

B. NER1,380,252.



B. NER1,380,252.



C. NER1,434,783.

Feedback

B is correct. Using the temporal method, monetary assets (i.e., cash) are translated using the current exchange rate (as of 31 December 2016) of BRD1.20/NER (or NER0.8333/BRD), and non-monetary assets are translated using the historical exchange rate when acquired. Inventory is translated at its 2016 weighted-average rate of BRD1.19/NER (or NER0.8403/BRD). Therefore, the total assets for Triofind-B translated into Norvolt euros (Triofind's presentation currency) as of 31 December 2016 are calculated as follows:

Assets	31 December 2016 (BRD)	Applicable Exchange Rate (NER/BRD)	Rate Used	NER
Cash	900,000	0.8333	Current	750,000
Inventory	750,000	0.8403	Average	630,252
Total	1,650,000			1,380,252



11 Multiple Choice 0 / 1 point

Based on Exhibits 1 and 2, the translation adjustment for Triofind-B's liabilities into Triofind's presentation currency for the six months ended 31 December 2016 is:



A.
negative.

Correct
Answer:

C. positive.



B. zero.



C. positive.

Feedback

C is correct. The monetary balance sheet items for Triofind-B are translated at the current exchange rate, which reflects that the Borliand dollar weakened during the period relative to the Norvolt euro. The rate as of 30 June 2016 was BRD1.15/NER (or NER/BRD0.8696) and as of 31 December 2016 was BRD1.20/NER (or NER/BRD0.8333). Therefore, notes payable translates to NER416,667 (BRD500,000 × NER/BRD0.8333) as of 31 December 2016, compared with NER434,783 (BRD500,000 × NER/BRD0.8696) as of 30 June 2016. Thus, the translation adjustment for liabilities is positive.

12 Multiple Choice 0 / 1 point

Based on Exhibits 1 and 2 and Janssen's translation method, retained earnings for Triofind-B translated into Triofind's presentation currency as of 30 June 2017 are *closest* to:

☐ A. NER150,225.

☒ B. NER170,940.

Correct Answer: **A. NER150,225.**

☐ C. NER172,414.

Feedback

A is correct. Triofind uses the temporal method to translate the financial statements of Triofind-B. The temporal method uses the current exchange rate for translating monetary assets and liabilities and the historical exchange rate (based on the date when the assets were acquired) for non-monetary assets and liabilities. Monetary assets and liabilities are translated using the current exchange rate (as of 30 June 2017) of NER1 = BRD1.17 (or NER0.8547/BRD), and non-monetary assets and liabilities are translated using the historical exchange rate (as of 30 June 2016) of NER1 = BRD1.15 (or NER0.8696/BRD). Inventory is translated at the 2017 weighted average rate of NER1 = BRD1.18 (or NER0.8475/BRD). The difference required to maintain equality between (a) total assets and (b) total liabilities and shareholder's equity is then recorded as retained earnings. The retained earnings for Triofind-B translated into Norvolt euros (Triofind's presentation currency) as of 30 June 2017 is calculated as follows:

	30 June 2017 (BRD)	Exchange Rate (NER/BRD)	Rate Used	30 June 2017 (NER)	Liabilities and Stockholders' Equity	30 June 2017 (BRD)
Assets						
Cash	1,350,000	0.8547	C	1,153,846	Notes Payable	500,000
Inventory	500,000	0.8475	H	423,729	Common Stock	1,150,000
					Retained Earnings	200,000
	1,850,000			1,577,575	Total	1,850,000

13 Multiple Choice 1 / 1 point

The functional currency for Triofind-A's sale of mobile phones to a customer in Certait is the:

☐ A. Certait real.

☐ B. Norvolt euro.

☒ C. Abuelio peso.

Feedback

C is correct. The functional currency is the currency of the primary economic environment in which an entity operates. Abuelio is Triofind-A's primary economic environment, and its currency is the Abuelio peso (ABP). Another important factor used to determine the functional currency is the currency that mainly influences sales prices for goods and services. The fact that Triofind-A prices its goods in Abuelio pesos supports the case for the ABP to be the functional currency.

14 Multiple Choice 1 / 1 point

Based on Exhibit 3, the value of the new warehouse in Abuelio on Triofind's balance sheet as of 31 July 2017 is *closest* to:

☐ A. NER31,452.

☒ B.
NER47,964.

☐ C. NER50,000.

Feedback

B is correct. Triofind complies with IFRS, and Abuelio can be considered a highly inflationary economy because its cumulative inflation rate exceeded 100% from 2015 to 2017. Thus, Triofind-A's financials must be restated to include local inflation rates and then translated using the current exchange rate into Norvolt euros, which is Triofind's presentation currency. This approach reflects both the likely change in the local currency value of the warehouse as well as the actual change in the exchange rate. The original purchase price is ABP1,008,065 (NER50,000/ABP0.0496). The value of the new warehouse in Abuelio as of 31 July 2017 is NER47,964, calculated as follows:

Date	Abuelio Monthly Inflation Rate (%)	Restated Warehouse Value (ABP)	NER/ABP	Warehouse Value (NER)
31 May 2017		1,008,065	0.0496	50,000
30 June 2017	25	1,260,081	0.0388	48,891
31 July 2017	22	1,537,298	0.0312	47,964

15 Multiple Choice 1 / 1 point

Relative to its domestic tax rate, Triofind's effective tax rate is *most likely*:

☒ A.
lower.

- ☐ B. the same.
- ☐ C. higher.

Feedback

A is correct. Norvolt exempts the non-domestic income of multinationals from taxation. Because Norvolt has a corporate tax rate of 34%, the 0% tax rate in Borliand and the fact that 25% of Triofind's net income comes from Borliand should result in a lower effective tax rate on Triofind's consolidated financial statements compared with Triofind's domestic tax rate. Abuelio's tax rate of 35% is very close to that of Norvolt, and it constitutes only 15% of Triofind's net income, so its effect is unlikely to be significant.

16 Multiple Choice 1 / 1 point

Based on Exhibit 4, the country with the highest sustainable sales growth is:

- ☐ A. Norvolt.
- ☒ B. Abuelio.
- ☐ C. Borliand.

Feedback

B is correct. Although Borliand shows the highest growth in Norvolt euro terms, this result is partially because of currency fluctuations, which cannot be controlled. Abuelio had the highest change in sales resulting from price and volume at 13% (excluding foreign currency exchange). This growth is more sustainable than net sales growth, which includes currency fluctuations, because Triofind's management has more control over growth in sales resulting from greater volume or higher prices.

Vignette

Pedro Ruiz is an analyst for a credit rating agency. One of the companies he follows, Eurexim SA, is based in France and complies with International Financial Reporting Standards (IFRS). Ruiz has learned that Eurexim used EUR220 million of its own cash and borrowed an equal amount to open a subsidiary in Ukraine. The funds were converted into hryvnia (UAH) on 31 December 20X1 at an exchange rate of EUR1.00 = UAH6.70 and used to purchase UAH1,500 million in fixed assets and UAH300 million of inventories.

Ruiz is concerned about the effect that the subsidiary's results might have on Eurexim's consolidated financial statements. He calls Eurexim's Chief Financial Officer, but learns little. Eurexim is not willing to share sales forecasts and has not even made a determination as to the subsidiary's functional currency.

Absent more useful information, Ruiz decides to explore various scenarios to determine the potential impact on Eurexim's consolidated financial statements. Ukraine is not currently in a hyperinflationary environment, but Ruiz is concerned that this situation could change. Ruiz also believes the euro will appreciate against the hryvnia for the foreseeable future.

17 Multiple Choice 0 / 1 point

If Ukraine's economy becomes highly inflationary, Eurexim will *most likely* translate inventory by:



☐ A. restating for inflation and using the temporal method.

Correct Answer:

B. restating for inflation and using the current exchange rate.

- ☐ B. restating for inflation and using the current exchange rate.
- ☐ C. using the temporal method with no restatement for inflation.

Feedback

B is correct. IAS 21 requires that the financial statements of the foreign entity first be restated for local inflation using the procedures outlined in IAS 29, "Financial Reporting in Hyperinflationary Economies." Then, the inflation-restated foreign currency financial statements are translated into the parent's presentation currency using the current exchange rate. Under US GAAP, the temporal method would be used with no restatement.

18 Multiple Choice 1 / 1 point

Given Ruiz's belief about the direction of exchange rates, Eurexim's gross profit margin would be *highest* if it accounts for the Ukraine subsidiary's inventory using:

☐ A. FIFO and the temporal method.



☒ B. FIFO and the current rate method.

☐ C. weighted-average cost and the temporal method.

Feedback

B is correct. Ruiz expects the EUR to appreciate against the UAH and expects some inflation in the Ukraine. In an inflationary environment, FIFO will generate a higher gross profit than weighted-average cost. For either inventory choice, the current rate method will give higher gross profit to the parent company if the subsidiary's currency is depreciating. Thus, using FIFO and translating using the current rate method will generate a higher gross profit for the parent company, Eurexim SA, than any other combination of choices.

19 Multiple Choice 1 / 1 point

If the euro is chosen as the Ukraine subsidiary's functional currency, Eurexim will translate its fixed assets using the:

☐ A. average rate for the reporting period.



☒ B. rate in effect when the assets were purchased.

☐ C. rate in effect at the end of the reporting period.

Feedback

B is correct. If the parent's currency is chosen as the functional currency, the temporal method must be used. Under the temporal method, fixed assets are translated using the rate in effect at the time the assets were acquired.

20 Multiple Choice 1 / 1 point

If the euro is chosen as the Ukraine subsidiary's functional currency, Eurexim will translate its accounts receivable using the:

- ☐ A. rate in effect at the transaction date.
- ☐ B. average rate for the reporting period.
- ☒ C. rate in effect at the end of the reporting period.

Feedback

C is correct. Monetary assets and liabilities such as accounts receivable are translated at current (end-of-period) rates regardless of whether the temporal or current rate method is used.

21 Multiple Choice 0 / 1 point

If the hryvnia is chosen as the Ukraine subsidiary's functional currency, Eurexim will translate its inventory using the:

- ☐ A. average rate for the reporting period.
- ☐ B. rate in effect at the end of the reporting period.
- ☒ C. rate in effect at the time the inventory was purchased.

Correct Answer:

B. rate in effect at the end of the reporting period.

Feedback

B is correct. When the foreign currency is chosen as the functional currency, the current rate method is used. All assets and liabilities are translated at the current (end-of-period) rate.

22 Multiple Choice 0 / 1 point

Based on the information available and Ruiz's expectations regarding exchange rates, if the hryvnia is chosen as the Ukraine subsidiary's functional currency, Eurexim will *most likely* report:

- ☒ A. an addition to the cumulative translation adjustment.
- ☐ B. a translation gain or loss as a component of net income.

Correct Answer:

C. a subtraction from the cumulative translation adjustment.

- ☐ C. a subtraction from the cumulative translation adjustment.

Feedback

C is correct. When the foreign currency is chosen as the functional currency, the current rate method must be used and all gains or losses from translation are reported as a cumulative translation adjustment to shareholder equity. When the foreign currency decreases in value (weakens), the current rate method results in a negative translation adjustment in stockholders' equity.

Vignette

Consolidated Motors is a US-based corporation that sells mechanical engines and components used by electric utilities. Its Canadian subsidiary, Consol-Can, operates solely in Canada. It was created on 31 December 20X1, and Consolidated Motors determined at that time that it should use the US dollar as its functional currency.

Chief Financial Officer Monica Templeton was asked to explain to the board of directors how exchange rates affect the financial statements of both Consol-Can and the consolidated financial statements of Consolidated Motors. For the presentation, Templeton collects Consol-Can's balance sheets for the years ended 20X1 and 20X2 (Exhibit 1), as well as relevant exchange rate information (Exhibit 2).

Exhibit 1

Consol-Can Condensed Balance Sheet for Fiscal Years Ending 31 December (C\$ millions)

Account	20X2	20X1
Cash	135	167
Accounts receivable	98	—
Inventory	77	30
Fixed assets	100	100
Accumulated depreciation	(10)	—
Total assets	400	297
Accounts payable	77	22
Long-term debt	175	175
Common stock	100	100
Retained earnings	48	—
Total liabilities and shareholders' equity	400	297

Exhibit 2

Exchange Rate Information

	US\$/C\$
Rate on 31 December 20X1	0.86
Average rate in 20X2	0.92
Weighted-average rate for inventory purchases	0.92
Rate on 31 December 20X2	0.95

Templeton explains that Consol-Can uses the FIFO inventory accounting method and that purchases of C\$300 million and the sell-through of that inventory occurred evenly throughout 20X2. Her presentation includes reporting the translated amounts in US dollars for each item, as well as associated translation-related gains and losses. The board responds with several questions.

- Would there be a reason to change the functional currency to the Canadian dollar?
- Would there be any translation effects for Consolidated Motors if the functional currency for Consol-Can were changed to the Canadian dollar?
- Would a change in the functional currency have any impact on financial statement ratios for the parent company?
- What would be the balance sheet exposure to translation effects if the functional currency were changed?

23 Multiple Choice 1 / 1 point

After translating Consol-Can's inventory and long-term debt into the parent company's currency (US\$), the amounts reported on Consolidated Motor's financial statements on 31 December 20X2 would be *closest* to (in millions):

- ☐ A. \$71 for inventory and \$161 for long-term debt.
- ☒ B. \$71 for inventory and \$166 for long-term debt.
- ☐ C. \$73 for inventory and \$166 for long-term debt.

Feedback

B is correct. When the parent company's currency is used as the functional currency, the temporal method must be used to translate the subsidiary's accounts. Under the temporal method, monetary assets and liabilities (e.g., debt) are translated at the current (year-end) rate, non-monetary assets and liabilities measured at historical cost (e.g., inventory) are translated at historical exchange rates, and non-monetary assets and liabilities measured at current value are translated at the exchange rate at the date when the current value was determined. Because beginning inventory was sold first and sales and purchases were evenly acquired, the average

rate is most appropriate for translating inventory and C\$77 million $\times$ 0.92 = \$71 million. Long-term debt is translated at the year-end rate of 0.95. C\$175 million $\times$ 0.95 = \$166 million.

24 Multiple Choice 0 / 1 point

After translating Consol-Can's 31 December 20X2 balance sheet into the parent company's currency (US\$), the translated value of retained earnings will be *closest* to:

☒ A. \$41 million.

Correct Answer: **B. \$44 million.**

☐ B. \$44 million.

☐ C. \$46 million.

Feedback

B is correct. Translating the 20X2 balance sheet using the temporal method, as is required in this instance, results in assets of US\$369 million. The translated liabilities and common stock are equal to US\$325 million, meaning that the value for 20X2 retained earnings is US\$369 million – US\$325 million = US\$44 million.

Temporal Method (20X2)

Account	C\$	Rate	US\$
Cash	135	0.95	128
Accounts receivable	98	0.95	93
Inventory	77	0.92	71
Fixed assets	100	0.86	86
Accumulated depreciation	(10)	0.86	(9)
Total assets	400		369
Accounts payable	77	0.95	73
Long-term debt	175	0.95	166
Common stock	100	0.86	86
Retained earnings	48	to balance	44
Total liabilities and shareholders' equity	400		369

25 Multiple Choice 1 / 1 point

In response to the board's first question, Templeton would *most likely* reply that such a change would be justified if:

- ☐ A. the inflation rate in the United States became hyperinflationary.
- ☐ B. management wanted to flow more of the gains through net income.
- ☒ C. Consol-Can were making autonomous decisions about operations, investing, and financing.

Feedback

C is correct. The Canadian dollar would be the appropriate reporting currency when substantially all operating, financing, and investing decisions are based on the local currency. The parent country's inflation rate is never relevant. Earnings manipulation is not justified, and at any rate changing the functional currency would take the gains off of the income statement.

26 Multiple Choice 0 / 1 point

In response to the board's second question, Templeton should reply that if the change is made, the consolidated financial statements for Consolidated Motors would begin to recognize:

- ☐ A. realized gains and losses on monetary assets and liabilities.
- ☒ B. realized gains and losses on non-monetary assets and liabilities.

Correct Answer: **C. unrealized gains and losses on non-monetary assets and liabilities.**

- ☐ C. unrealized gains and losses on non-monetary assets and liabilities.

Feedback

C is correct. If the functional currency were changed from the parent currency (US dollar) to the local currency (Canadian dollar), the current rate method would replace the temporal method. The temporal method ignores unrealized gains and losses on non-monetary assets and liabilities, but the current rate method does not.

27 Multiple Choice 1 / 1 point

In response to the board's third question, Templeton should note that the change will *most likely* affect:

- ☐ A. the cash ratio.
- ☒ B. fixed asset turnover.

- ☐ C. receivables turnover.

Feedback

B is correct. If the Canadian dollar is chosen as the functional currency, the current rate method will be used and the current exchange rate will be the rate used to translate all assets and

liabilities. Currently, only monetary assets and liabilities are translated at the current rate. Sales are translated at the average rate during the year under either method. Fixed assets are translated using the historical rate under the temporal method but would switch to current rates under the current rate method. Therefore, there will most likely be an effect on sales/fixed assets. Because the cash ratio involves only monetary assets and liabilities, it is unaffected by the translation method. Receivables turnover pairs a monetary asset with sales and is thus also unaffected.

28 Multiple Choice 0 / 1 point

In response to the board's fourth question, the balance sheet exposure (in C\$ millions) would be *closest* to:

☒ A. -19.

Correct Answer: **B. 148.**

☐ B. 148.

☐ C. 400.

Feedback

B is correct. If the functional currency were changed, then Consol-Can would use the current rate method and the balance sheet exposure would be equal to net assets (total assets – total liabilities). In this case, $400 - 77 - 175 = 148$.

Vignette

Romulus Corp. is a US-based company that prepares its financial statements in accordance with US GAAP. Romulus Corp. has two European subsidiaries: Julius and Augustus. Anthony Marks, CFA, is an analyst trying to forecast Romulus's 20X2 results. Marks has prepared separate forecasts for both Julius and Augustus, as well as for Romulus's other operations (prior to consolidating the results.) He is now considering the impact of currency translation on the results of both the subsidiaries and the parent company's consolidated financials. His research has provided the following insights:

- The results for Julius will be translated into US dollars using the current rate method.
- The results for Augustus will be translated into US dollars using the temporal method.
- Both Julius and Augustus use the FIFO method to account for inventory.
- Julius had year-end 20X1 inventory of €340 million. Marks believes Julius will report €2,300 in sales and €1,400 in cost of sales in 20X2.

Marks also forecasts the 20X2 year-end balance sheet for Julius (Exhibit 1). Data and forecasts related to euro/dollar exchange rates are presented in Exhibit 2.

Exhibit 1

Forecasted Balance Sheet Data for Julius, 31 December 20X2 (€ millions)

Cash	50
Accounts receivable	100
Inventory	700
Fixed assets	1,450
Total assets	2,300
Liabilities	700
Common stock	1,500
Retained earnings	100
Total liabilities and shareholder equity	2,300

Exhibit 2 Exchange Rates (\$/€)

31 December 20X1	1.47
31 December 20X2	1.61
20X2 average	1.54
Rate when fixed assets were acquired	1.25
Rate when 20X1 inventory was acquired	1.39
Rate when 20X2 inventory was acquired	1.49

29 Multiple Choice 1 / 1 point

Based on the translation method being used for Julius, the subsidiary is *most likely*:

- ☐ A. a sales outlet for Romulus's products.
- ☒ B. a self-contained, independent operating entity.
- ☐ C. using the US dollar as its functional currency.

Feedback

B is correct. Julius is using the current rate method, which is most appropriate when it is operating with a high degree of autonomy.

30 Multiple Choice 1 / 1 point

To account for its foreign operations, Romulus has *most likely* designated the euro as the functional currency for:



☒ A. Julius only.

☐ B. Augustus only.

☐ C. both Julius and Augustus.

Feedback

A is correct. If the current rate method is being used (as it is for Julius), the local currency (euro) is the functional currency. When the temporal method is being used (as it is for Augustus), the parent company's currency (US dollar) is the functional currency.

31 Multiple Choice 1 / 1 point

When Romulus consolidates the results of Julius, any unrealized exchange rate holding gains on monetary assets should be:

☐ A. reported as part of operating income.

☐ B. reported as a non-operating item on the income statement.



☒ C. reported directly to equity as part of the cumulative translation adjustment.

Feedback

C is correct. When the current rate method is being used, all currency gains and losses are recorded as a cumulative translation adjustment to shareholder equity.

32 Multiple Choice 0 / 1 point

When Marks translates his forecasted balance sheet for Julius into US dollars, total assets as of 31 December 20X2 (dollars in millions) will be *closest* to:



☐ A. \$1,429.

Correct Answer:

C. \$3,703.

- ☐ B. \$2,392.
- ☐ C. \$3,703.

Feedback

C is correct. Under the current rate method, all assets are translated using the year-end 20X2 (current) rate of \$1.61/€1.00. $€2,300 \times 1.61 = \$3,703$.

33 Multiple Choice 1 / 1 point

When Marks converts his forecasted income statement data for Julius into US dollars, the 20X2 gross profit margin will be *closest* to:

☒ A. 39.1%.

☐ B. 40.9%.

☐ C. 44.6%.

Feedback

A is correct. Under the current rate method, both sales and cost of goods sold would be translated at the 20X2 average exchange rate. The ratio would be the same as reported under the euro. $€2,300 - €1,400 = €900$, $€900/€2,300 = 39.1\%$. Or, $\$3,542 - \$2,156 = \$1,386$, $\$1,386/\$3,542 = 39.1\%$.

34 Multiple Choice 1 / 1 point

Relative to the gross margins the subsidiaries report in local currency, Romulus's consolidated gross margin *most likely*:

☐ A. will not be distorted by currency translations.

☐ B. would be distorted if Augustus were using the same translation method as Julius.

☒ C. will be distorted because of the translation and inventory accounting methods Augustus is using.

Feedback

C is correct. Augustus is using the temporal method in conjunction with FIFO inventory accounting. If FIFO is used, ending inventory is assumed to be composed of the most recently acquired items, and thus inventory will be translated at relatively recent exchange rates. To the extent that the average weight used to translate sales differs from the historical rate used to translate inventories, the gross margin will be distorted when translated into US dollars.

Vignette

Redline Products, Inc. is a US-based multinational with subsidiaries around the world. One such subsidiary, Acceletron, operates in Singapore, which has seen mild but not excessive rates of inflation. Acceletron was acquired in 2000 and has never paid a dividend. It records inventory using the FIFO method.

Chief Financial Officer Margot Villiers was asked by Redline's board of directors to explain how the functional currency selection and other accounting choices affect Redline's consolidated financial statements. Villiers gathers Acceletron's financial statements denominated in Singapore dollars (SGD) in Exhibit 1 and the US dollar/Singapore dollar exchange rates in Exhibit 2. She does not intend to identify the functional currency actually in use but rather to use Acceletron as an example of how the choice of functional currency affects the consolidated statements.

Exhibit 1

Selected Financial Data for Acceletron, 31 December 2007 (SGD millions)

Cash	SGD125
Accounts receivable	230
Inventory	500
Fixed assets	1,640
Accumulated depreciation	(205)
Total assets	SGD2,290
Accounts payable	185
Long-term debt	200
Common stock	620
Retained earnings	1,285
Total liabilities and equity	2,290
Total revenues	SGD4,800
Net income	SGD450

Exhibit 2

Exchange Rates Applicable to Acceletron

Exchange Rate in Effect at Specific Times**USD per SGD**

Rate when first SGD1 billion of fixed assets were acquired	0.568
Rate when remaining SGD640 million of fixed assets were acquired	0.606
Rate when long-term debt was issued	0.588
31 December 2006	0.649
Weighted-average rate when inventory was acquired	0.654
Average rate in 2007	0.662
31 December 2007	0.671

35 Multiple Choice 0 / 1 point

Compared with using the Singapore dollar as Acceletron's functional currency for 2007, if the US dollar were the functional currency, it is *most likely* that Redline's consolidated:

- ☐ A. inventories will be higher.
- ☒ B. receivable turnover will be lower.

Correct Answer: **C. fixed asset turnover will be higher.**

- ☐ C. fixed asset turnover will be higher.

Feedback

C is correct. If the US dollar is the functional currency, the temporal method must be used. Revenues and receivables (monetary asset) would be the same under either accounting method. Inventory and fixed assets were purchased when the US dollar was stronger, so at historical rates (temporal method), translated they would be lower. Identical revenues/lower fixed assets would result in higher fixed-asset turnover.

36 Multiple Choice 0 / 1 point

If the US dollar were chosen as the functional currency for Acceletron in 2007, Redline could reduce its balance sheet exposure to exchange rates by:

- ☐ A. selling SGD30 million of fixed assets for cash.
- ☒ B. issuing SGD30 million of long-term debt to buy fixed assets.

Correct Answer: **A. selling SGD30 million of fixed assets for cash.**

- ☐ C. issuing SGD30 million in short-term debt to purchase marketable securities.

Feedback

A is correct. If the US dollar is the functional currency, the temporal method must be used, and the balance sheet exposure will be the net monetary assets of $125 + 230 - 185 - 200 = -30$, or a net monetary liability of SGD30 million. This net monetary liability would be eliminated if fixed assets (non-monetary) were sold to increase cash. Issuing debt, either short-term or long-term, would increase the net monetary liability.

37 Multiple Choice 1 / 1 point

Redline's consolidated gross profit margin for 2007 would be *highest* if Acceletron accounted for inventory using:



☒ A. FIFO, and its functional currency were the US dollar.

☐ B. LIFO, and its functional currency were the US dollar.

☐ C. FIFO, and its functional currency were the Singapore dollar.

Feedback

A is correct. Because the US dollar has been consistently weakening against the Singapore dollar, cost of sales will be lower and gross profit higher when an earlier exchange rate is used to translate inventory, compared with using current exchange rates. If the Singapore dollar is the functional currency, current rates would be used. Therefore, the combination of the US dollar (temporal method) and FIFO will result in the highest gross profit margin.

38 Multiple Choice 0 / 1 point

If the current rate method is used to translate Acceletron's financial statements into US dollars, Redline's consolidated financial statements will *most likely* include Acceletron's:

☐ A. USD3,178 million in revenues.

☐ B. USD118 million in long-term debt.



☐ C. negative translation adjustment to shareholder equity.

Correct Answer:

A. USD3,178 million in revenues.

Feedback

A is correct. Under the current rate method, revenue is translated at the average rate for the year, $\text{SGD}4,800 \times 0.662 = \text{USD}3,178$ million. Debt should be translated at the current rate, $\text{SGD}200 \times 0.671 = \text{USD}134$ million. Under the current rate method, Acceletron would have a net asset balance sheet exposure. Because the Singapore dollar has been strengthening against the US dollar, the translation adjustment would be positive rather than negative.

39 Multiple Choice 1 / 1 point

If Acceletron's financial statements are translated into US dollars using the temporal method, Redline's consolidated financial statements will *most likely* include Acceletron's:

- ☐ A. USD336 million in inventory.
- ☒ B. USD956 million in fixed assets.

- ☐ C. USD152 million in accounts receivable.

Feedback

B is correct. Under the temporal method, inventory and fixed assets would be translated using historical rates. Accounts receivable is a monetary asset and would be translated at year-end (current) rates. Fixed assets are found as $(1,000 \times 0.568) + (640 \times 0.606) = \text{USD } 956 \text{ million}$.

40 Multiple Choice 1 / 1 point

When translating Acceletron's financial statements into US dollars, Redline is *least likely* to use an exchange rate of USD per SGD:

- ☐ A. 0.671.
- ☒ B. 0.588.
- ☐ C. 0.654.

Feedback

B is correct. USD0.671/SGD is the current exchange rate. That rate would be used regardless of whether Acceletron uses the current rate or temporal method. USD0.654 was the weighted-average rate when inventory was acquired. That rate would be used if the company translated its statements under the temporal method but not the current rate method. USD0.588/SGD was the exchange rate in effect when long-term debt was issued. As a monetary liability, long-term debt is always translated using current exchange rates. Consequently, that rate is not applicable regardless of how Acceletron translates its financial statements.

Vignette

Mark Crawley is an analyst at a London-based private equity firm and is reviewing the firm's file on Thames Air Plc (Thames), a company it provides financing for. Thames uses International Financial Reporting Standards (IFRS) in the preparation of its financial statements. Thames is a relatively new airline based in the United Kingdom specializing in flights and vacation packages to Mediterranean locations, primarily Spain. Thames sells most of its flights and vacation packages to British residents in British pounds (GBP) and considers the costs of local competitors' packages when determining its prices. Costs are incurred in multiple currencies:

- Wage costs are primarily in GBP.
- Typical of the industry, airline fuel and lease costs are normally priced in US dollars (USD).
- The landing fees paid at the vacation-area airports are in the local currency, primarily euros (EUR).

First, Crawley turns his attention to the effect of the transactions undertaken in various currencies by Thames.

- He reviews the change in the exchange rate for the USD to GBP during 2015, shown in Exhibit 1, and wonders what the effect of this change was on Thames's operating income.
- At year-end (31 December), Thames had a large outstanding payable in Spain related to landing fees that were incurred there evenly over the final quarter. The company paid the amount in full on its due date of 28 February. Crawley observed that the EUR to GBP exchange rate had changed between when the costs were incurred and the year-end and again by the payment date, as also shown in Exhibit 1.

Exhibit 1 Selected Exchange Rate Data

	GBP/USD Close	GBP/EUR Close
1-Jan-15	0.64	---
30-Jun-15	---	0.72
31-Dec-15	0.68	0.75
28-Feb-16	---	0.73
Average, 1 July–31 December 2015	---	0.7325
Average, 1 October–31 December 2015	---	0.74

Because of the growing demand for vacation rentals in Spain during the past year, Thames acquired 100% of Tagus SA (Tagus), a Spanish company that owns a small vacation hotel and a few villas. Tagus has long-term debt outstanding from a Spanish bank that financed the 2012 purchase of the vacation properties, which will now be rented as part of the vacation packages offered by Thames. Tagus incurs all costs related to operating and maintaining the rental properties in EUR.

Since the acquisition, all of Tagus's revenue comes from Thames's sales in Britain of the vacation packages. Tagus receives the amounts in GBP. But Tagus hopes to expand and start renting out any excess capacity of the properties, or newly acquired properties, to local tourists in the next few years. Crawley notices that Thames is using the temporal method to translate Tagus's financial statements prior to consolidation and asks another analyst, Dee Chopra, if this is appropriate.

Crawley next reviews the information in Exhibit 2 related to the Tagus acquisition to consider the effect on Thames's year-end financial statements (31 December 2015).

Exhibit 2 Selected Financial Information of Tagus SA at Acquisition and Year-End (EUR thousands)

	Balance Sheet		Income Statement	
	Date of Acquisition (30 June 2015)	Year-End (31 December 2015)	Six-Month Period Ending 31 December 2015	
Cash and accounts receivable	4,000	4,200	Revenues	8,200
Inventory	2,000	2,250	Operating costs	6,485
Capital assets	15,000	14,625	Operating income	1,715
Total assets	21,000	21,075	Interest expense	395
			Earnings before taxes	1,320
Current liabilities	3,500	3,400	Income taxes	395
Long-term debt	10,000	9,750	Earnings after tax	925
Share capital	5,000	5,000		
			Dividends	500
Retained earnings	2,500	2,925		

As the final step in his review, Crawley starts a ratio analysis of Thames and Tagus, and he asks Chopra which ratios, if any, would be unaffected by Thames's choice of translation method for Tagus.

41 Multiple Choice 0 / 1 point

The functional currency of Thames is *most likely*:

☐ EUR.

Incorrect Answer Feedback: Incorrect. Although Thames incurs landing fees in euros and it sub operates in the Eurozone, there are more factors that support the pound as the functional currency of Thames.

☐ GBP.

Correct Answer Feedback: Correct. The functional currency is the currency of the primary economic environment in which an entity operates. Using the IFRS factors for determining the functional currency, Thames's functional currency is most likely GBP. The company sets its prices in GBP based on competitive factors in the British market; it is financed, to date, by a British private equity firm, and it receives its revenues primarily in GBP.

☒ USD.

Incorrect Answer Feedback: Incorrect. Landing fees and aircraft costs would be a large portion of Thames' costs, but operating revenues and financing are in pounds, and the IFRS list of factors gives priority to those.

42 Multiple Choice 1 / 1 point

Which of the following statements about the effect of the change in the USD to GBP exchange rate during the year is *most* accurate? Operating income for Thames would:

☐ increase because of the positive effect on revenues.

Incorrect Answer Feedback: Incorrect. The strengthening USD would not significantly affect revenues because most sales are in GBP.

☐ increase because of the positive effect on operating costs.

Incorrect Answer Feedback: Incorrect. The USD strengthened thereby increasing the costs incurred in USD and decreasing operating margins.

☒ decrease because of the negative effect on operating costs.

Correct Answer Feedback: Correct. Thames incurs fuel and lease costs in USD. During the year, the USD has strengthened relative to the GBP (from GBP0.6400/USD to GBP0.6800/USD), thereby making purchases (costs) in USD more expensive for Thames and thus decreasing operating income.

43 Multiple Choice 0 / 1 point

Which of the following *best* describes the effect on Thames's financial statements of the payment terms related to the landing fees in Spain? Thames would:

☐ report an unrealized exchange loss at year-end.

Correct Answer Feedback: Correct. Between incurring the fees and the year-end, the EUR appreciated against the GBP (GBP0.7400/EUR to GBP0.7500/EUR), and Thames would report an unrealized loss (from holding a liability in an appreciating currency) in net income. The payable would be originally recorded when the landing fees were incurred during the final quarter (GBP0.7400) but revalued at year-end to GBP0.7500; an increase in a liability results in a loss.

☐ defer recognizing any currency effects until the payable is paid.

Incorrect Answer Feedback: Incorrect. FX gains/losses from payables (monetary liabilities) are reported in the period they occur even if not realized.

☒ adjust the landing fees expense to reflect the change in exchange rate when they are paid.

Incorrect Answer Feedback: Incorrect. The expense would not be adjusted. The choice to not pay it immediately and incur a payable is a financing decision, and the gains and losses from that are not adjusted to the operating cost.

44 Multiple Choice 0 / 1 point

Chopra's *best* answer to Crawley's question about Thames's use of the temporal method to translate Tagus's financial statements is that it is:

- ☐ correct, because the functional currency of Tagus is GBP.

Correct Answer Feedback: Correct. The functional currency of Tagus currently appears to be GBP. The operations of Tagus are presently just an extension of Thames as supported by the fact that the only source of revenue is payments from Thames, received in GBP. Even though Tagus incurs most costs and its financing (long-term debt) in EUR, under IFRS, Factors 1 and 2 (dealing with revenues and sales prices) rank higher than Factors 3 and 4 (dealing with operating and financing costs) when determining functional currency. Under IFRS, when the functional currency of the subsidiary is the same as the parent, the temporal method should be used to translate the subsidiary's financial statements.

- ☒ incorrect, because the functional currency of Tagus is EUR.

Incorrect Answer Feedback: Incorrect because the functional currency of Tagus is the pound not the euro, despite incurring most costs and their financing in the euro. Factors 1 and 2 (dealing with revenues) rank higher than Factors 3 and 4 (dealing with operating and financing costs) in the IFRS hierarchy of factors determining functional currency.

- ☐ correct, if the presentation currency of Tagus's financial statements is GBP.

Incorrect Answer Feedback: Incorrect because it is the subsidiary's functional currency that determines the translation method not the subsidiary's presentation currency.

45 Multiple Choice 1 / 1 point

The *most likely* effect of the change in the exchange rate between the EUR and GBP arising from Thames's investment in Tagus in 2015 will be a translation:

- ☒ loss reported in net income.

Correct Answer Feedback: Correct. Thames is using the temporal method for its translation of Tagus, and the initial exposure is a net liability exposure (monetary liabilities of EUR13,500 exceed the monetary assets of EUR4,000). The EUR strengthened against the GBP during the six-month period (from GBP0.7200/EUR to GBP0.7500/EUR). The net effect of having a net liability position in a strengthening currency will be a translation loss for Thames, which, under the temporal method, is reported in net income.

- ☐ gain reported in net income.

Incorrect Answer Feedback: Incorrect. Due to the net liability position in a strengthening currency, it will be a loss not a gain.

- ☐ adjustment reported in other comprehensive income.

Incorrect Answer Feedback: Incorrect. Under the temporal method, it is reported in net income not other comprehensive income.

46 Multiple Choice 1 / 1 point

The *best* answer Chopra can give to Crawley's question about which ratio would be unaffected is the:

- ☐ current ratio.

Incorrect Answer Feedback: Incorrect. Under the temporal method, the monetary (cash and A/R) and non-monetary (inventory) components of current assets are translated at different rates (balance sheet rate and historical respectively) vs. under the current rate method, where all are translated at the balance sheet rate. Therefore, the ratio will differ under the two methods because the total current assets once translated would differ.

- ☐ operating profit margin.

Incorrect Answer Feedback: Incorrect. The sales are translated at the same rate (average) under the temporal and current methods, but the costs of goods sold are not. Under the current method they are at average, but under the temporal method it depends on when the inventory was purchased. Therefore, the ratios will differ under the two methods.

receivables turnover ratio.

Correct Answer Feedback: Correct. The accounts receivable turnover ratio is calculated as sales/average accounts receivable. Under both the current method and the temporal method, both sales and accounts receivable are translated at the same rates: average for sales and year-end for accounts receivable (accounts receivable are a monetary asset for the temporal method). Therefore, this ratio would not be affected by the method of translation.

Vignette

Nation Resorts Ltd (Nation) is a US-based operator of destination vacation resorts. Resorts are divided into three types: Winter, which are located at ski mountains; Golf, which are located on championship courses in such locations as Arizona; and Seaside, which are located at world-class ocean beaches. The three types of resorts generally attract different types of vacationers and provide different vacation packages for different budgets. Although the Winter and Golf resorts are somewhat counter seasonal, the company still suffers from risks related to the US economy, as well as the risk of unfavorable local weather conditions (lack of snow or sun). To diversify and reduce these risks, Nation started purchasing properties in other geographic areas, including Europe and Central America.

Nation made its first non-domestic investment when it bought a resort in Jamaica on 1 May 2008. Initially, Nation provided managers to assist in operating the resort but then hired local Jamaicans into its management development program, one of whom was recently appointed to a senior position at the resort. Nation invested in building and upgrading facilities, with the initial expansion financing provided by Nation's US bank. Concerned about the high rate of inflation in Jamaica, which at the time of purchase averaged 20% per year over the previous three years, Nation kept any excess funds in US dollars. The company's vacation packages are sold primarily in the United States, and its prices reflect competitive conditions in the US vacation market. In 2010, the Jamaican resort used local labor and supplies and was profitable for the first time. In addition, by that time, inflation had declined to 14% per year, and Nation expected to be able to reinvest any profits in the resort and start using a local bank for ongoing financing needs.

On 1 July 2010, Nation acquired its first European resort, Val Blanc SA, a ski resort in the Alps region of France. Nation paid €28 million for 100% of the company. The resort attracts skiers primarily from France and other European countries. The transition went very well, and Nation left the local managers in place to make all operating and financial decisions. By the end of 2011, Nation's only financial contribution to the resort was the initial equity investment. Exhibit 1 shows financial statements for the ski resort at acquisition and for the following six months.

Exhibit 1 Val Blanc Financial Statements (€ thousands)

	Income Statement	
	31 Dec. 2010 (6 months)	30 Jun. 2010 (12 months)
Resort revenues	12,025	46,805
Operating expenses	9,800	35,105
Depreciation and amortization	1,750	4,000
Interest expense	1,200	2,500
Earnings before taxes	-725	5,200
Income taxes	0	1,300
Net earnings	-725	3,900
	Statement of Financial Position	
	31 Dec. 2010	30 Jun. 2010 ^a
Cash and marketable securities	5,750	5,000
Accounts receivable	6,500	3,000
Inventory	8,000 ^b	5,000
Total current assets	20,250	13,000
Property, plant, and equipment	41,750	43,500
Intangible assets	5,000	5,000

Total assets	67,000	61,500
Current liabilities	12,225	5,000
Long-term debt	26,000	27,000
Share capital	20,000	20,000
Retained earnings	8,775	9,500
Total liabilities and equity	67,000	61,500
^a The 30 June 2010 Statement of Financial Position reflects the fair value of the identifiable assets and liabilities of Val Blanc at acquisition.		
^b The 31 December 2010 inventory was acquired evenly throughout the period since acquisition.		

In early January 2011, Paul Nakiska, a business analyst with Nation, met with Nation's manager of financial reporting, Max Chara, to discuss how the company should account for the two non-domestic resorts in the 2010 financial statements and the impact the resorts would have on Nation's reported results. Nation has a 31 December year end and prepares its financial statements according to US GAAP.

In preparation for the meeting, Nakiska's first task was to prepare the purchase price allocation of the Val Blanc acquisition using the acquisition method. He also gathered some exchange rate information related to the two resorts, shown in Exhibit 2.

Exhibit 2 Selected Exchange Rates

Date	USD/JMD	USD/EUR
5/1/2008	0.0139	N/A
1/1/2010	0.0115	1.4368
6/30/2010	0.0117	1.225
12/31/2010	0.0117	1.3261
Average 2010	0.0115	1.3277
Average January–July 2010	0.0113	1.3303
Average July–December 2010	0.0117	1.3198

Nakiska started the meeting:

"I suggest we use the current rate method for both our non-domestic subsidiaries because that will simplify our financial reporting. The current rate method also allows the key metrics we use to evaluate performance—the current ratio, fixed asset turnover, and operating margin—to be the same after translation as they are before."

Chara replied:

"I am not so concerned about the translated metrics because we evaluate the performance of each resort in its local currency. I am concerned, however, about the effect on Nation's consolidated net income and return on equity.

If we use the temporal method for the resort in France, we can take advantage of the strengthening euro and report the translation gain on the income statement.

I am also not concerned about using the same method for all subsidiaries if using different methods will help increase our net income.

Nakiska agreed to calculate the effects of the various translation methods and of the Val Blanc acquisition on the financial statements and send a report to Chara.

47 Multiple Choice 1 / 1 point

In 2008, Nation *most likely* used which of the following translation methods for the Jamaican resort?

☐ The temporal method because of the high rate of inflation

Incorrect Answer Feedback: Incorrect. The temporal method would be required under US GAAP if it were hyperinflationary (more than 100% over three years).

- ☒ The temporal method because the US dollar was the functional currency

Correct Answer Feedback: Correct. Based on the facts described, the US dollar was the Jamaican resort's functional currency in 2008. The US economy and the US dollar are the factors that influence the prices of the vacation packages. The operations were financed from the United States, and excess funds were invested there. Nation also supplied onsite management. Therefore, the temporal method would be the most appropriate method to use. Although inflation was high at 20% per year over the prior three years, it did not meet the US GAAP definition of a highly inflationary economy (a cumulative three-year inflation rate exceeding 100%), which requires the temporal method be used for subsidiaries normally using the current rate method during periods of hyperinflation.

- ☐ The current rate method because the Jamaican dollar was the functional currency

Incorrect Answer Feedback: Incorrect. Even though some of the factors to determine the functional currency are mixed, priority is given to the determination of prices, so the US dollar is the functional currency, not the Jamaican dollar.

48 Multiple Choice 0 / 1 point

The net income or loss (in thousands) from the Val Blanc subsidiary that was included in Nation's income for 2010 is *closest* to:

- ☒ -\$961.

Incorrect Answer Feedback: Incorrect. It uses the year-end rate instead of the average rate $(725) \times 1.3261 = (961)$. Also, if one uses the average rate for the year instead of for just the 6 months, $725 \times 1.3277 = 962.6$, which is close to 961.

- ☐ -\$957.

Correct Answer Feedback: Correct. The functional currency for the Val Blanc subsidiary is the euro because the resort operates independently and attracts European tourists. The prices are set in euros, and the resort is also financed in euros. Therefore, the financial statements would be translated using the current rate method. Under the current rate method, the net earnings (loss) is calculated using the average rate for the July–December 2010 period: $\text{€}725 \text{ thousand loss} \times \$1.3198/\text{€}1 = \text{\$}957 \text{ thousand}$.

- ☐ -\$791.

Incorrect Answer Feedback: Incorrect. It incorrectly converts depreciation at the historical rate (which is the temporal method).

	Euros	Rate	\$
Resort revenues	12,025	1.3198	15,871
Operating expenses	9,800	1.3198	12,934
Depreciation and amortization	1,750	1.225	2,144
Interest expense	1,200	1.3198	1,584
Earnings before taxes	(725)	---	(791)
Income taxes	0	---	---
Net earnings	(725)	---	---

49 Multiple Choice 0 / 1 point

Using the translation method suggested by Nakiska, the total shareholders' equity (in thousands) of the Val Blanc resort on 31 December 2010 after translation is *closest* to:

- ☐ \$32,203.

Incorrect Answer Feedback: Incorrect. It includes the translation adjustment as a loss not a gain: $35,181 - 2,978 = 32,203$.

- ☒ \$35,181.

Incorrect Answer Feedback: Incorrect. It does not include the translation gain/loss but only the share capital and retained earnings at correct rates: 35,181, from the last table above.

☐ \$38,159.

Correct Answer Feedback: Correct. Total shareholders' equity as of 31 December 2010 consists of share capital, retained earnings, and the cumulative foreign exchange gain or loss since acquisition. The total amount can be calculated either as the sum of the individual components or as the difference between assets and liabilities. Using the current rate method, all assets and liabilities are translated at the closing rate.

Method 1: Using the difference between assets and liabilities.

(in thousands)	EUR	USD/EUR	USD
Cash and marketable securities	5,750	1.3261	7,625
Accounts receivable	6,500	1.3261	8,620
Inventory	8,000	1.3261	10,609
Total current assets	20,250	---	---
Property, plant, and equipment	41,750	1.3261	55,365
Intangible assets	5,000	1.3261	6,630
Total assets	67,000	Total assets	88,849
Current liabilities	12,225	1.3261	16,211
Long-term debt	26,000	1.3261	34,479
Share capital	20,000	Total liabilities	50,690
Retained earnings	8,775	Net assets	38,159
Total liabilities and equity	67,000	---	---

Alternatively: $1.3261 \times [67,000 - (12,225 + 26,000)] = 38,159$

Method 2: Using the translation of individual items and calculation of the foreign exchange gain.

(in thousands)	EUR	USD/EUR	USD
Share capital	20,000	1.225	24,500
Retained earnings – opening	9,500	1.225	11,638
Net earnings (loss) for the period	(725)	1.3198	(957)
Foreign translation adjustment (see following calculation)	---	---	2,978
Total shareholders' equity	---	---	38,159

In the current rate method, the share capital and opening retained earnings use the rate at the date of acquisition (1 July), and the change in retained earnings is equal to the net earnings (loss) for the period at the average rate (July–December).

Foreign Translation Gain/Loss Calculation:

	EUR	USD/EUR	USD
Opening net asset position	29,500	1.225	36,138
Net loss	(725)	1.3198	(957)
Closing net asset position	28,775	---	35,181
Actual net asset position	28,775	1.3261	38,159
Gain (Loss)	---	---	2,978

Multiple Choice 0 / 1 point

Nakiska's comment about the key performance metrics is *least* accurate with respect to the:

☒ current ratio.

Incorrect Answer Feedback: Incorrect. All assets and liabilities are translated at the closing rate. Therefore, all ratios involving only asset and liability numbers, like the current ratio, will be unaffected by translation.

- ☐ operating margin.

Incorrect Answer Feedback: Incorrect. Under the current rate all revenues and expenses are translated at the average rate for the year, so ratios only involving income statement numbers, like the operation margin, will be unaffected by translation.

- ☐ fixed asset turnover.

Correct Answer Feedback: Correct.

Ratios involving:	Effect on financial statement and ratios under the current rate method
Only income statement items	All revenues and expenses are translated at the average rate for the year. Therefore, ratios involving only income statement numbers, such as the operating margin, will be unaffected by translation.
Balance sheet items (other than equity)	All assets and liabilities are translated at the closing rate. Therefore, all ratios involving only asset and liability numbers, such as the current ratio, will be unaffected by translation.
Both income statement and balance sheet items	Because the two financial statements are translated at different rates, the fixed asset turnover ratio (Sales/Fixed assets) is unlikely to provide the same result before and after translation, unless by chance the average rate happens to equal the closing rate.

51 Multiple Choice 0 / 1 point

Following Chara's assessment of the effect of using the temporal method on the Val Blanc resort, Nakiska's *most* appropriate conclusion is that Chara is:

- ☒ correct, because it would result in a translation gain on the income statement.

Incorrect Answer Feedback: Incorrect. Under the current method it would be a gain (an exposed net asset position with a strengthening currency), and it would go to other comprehensive income, not net income.

- ☐ incorrect, because it would result in a translation loss on the income statement.

Correct Answer Feedback: Correct. The euro strengthened against the US dollar in the July–December period (USD1.225/EUR to USD1.3261/EUR). Under the temporal method, the exposure is limited to monetary assets less monetary liabilities, which normally results in a net liability position (and does here for the Val Blanc subsidiary). When the non-domestic currency strengthens, a net liability position results in a negative translation adjustment (a loss). Hence, use of the temporal method would result in a translation loss being reported on the income statement, so Chara's statement is incorrect.

- ☐ incorrect, because the translation gain or loss would not be reported in net income.

Incorrect Answer Feedback: Incorrect. Under the current method it would be a gain (an exposed net asset position with a strengthening currency), and it would go to other comprehensive income, not net income.

Vignette

Claus Petersen, a pension fund equity analyst, is preparing an analysis of Rhine AG for the upcoming quarterly fund meeting. Rhine is a Germany-based manufacturer that operates three distinct divisions: children's products (infant car seats, strollers, cribs, etc.), recreational products (bicycles, bicycle trailers, etc.), and home furnishings (contemporary furniture). All three divisions sell through retail outlets around the world.

The company has been pursuing an aggressive growth strategy, achieved through both foreign acquisitions and organic growth. Petersen is interested in determining how well Rhine is allocating its resources between the three divisions and the effects of the foreign acquisitions on overall performance. Exhibit 1 summarizes selected divisional and corporate data for the current year and the prior year.

Exhibit 1 Rhine AG Selected Divisional and Corporate Data (€ millions)

	Total for Three Divisions		Children's Products		Recreational Products		Home Furnishings	
	Current Year	Prior Year	Current Year	Prior Year	Current Year	Prior Year	Current Year	Prior Year
Revenues	2,837.1	2,775.5	1,176.2	1,236.2	1,034.1	930.0	626.8	609.3
Gross profit	621.4	640.8	296.6	337.6	246.0	220.3	78.8	82.9
Operating profit	172.7	219.4	64.7	115.7	72.9	62.2	35.1	41.5
Earnings before taxes (EBT)	136.6	170.0	---	---	---	---	---	---
Net earnings after tax	109.9	132.3	---	---	---	---	---	---
Total assets	2,498.0	2,479.5	1,270.9	1,249.6	961.5	948.5	265.6	281.4
Capital expenditures	32.7	42.3	22.1	30.0	6.7	8.6	3.9	3.7
Proportion of capital expenditures	100%	100%	67.6%	70.9%	20.5%	20.3%	11.9%	8.7%
Proportion of total assets	100%	100%	50.9%	50.4%	38.5%	38.3%	10.6%	11.3%

Petersen's preferred method to determine which division is becoming less significant over time is to review the relationship between capital expenditures and total assets by operating division. He plans to base his conclusion on the assumption that the current year's investment behavior is representative of future investment patterns.

Petersen knows that revenues in the children's products division have suffered because of declining birth rates in Europe and North America, but he believes that if Rhine can maintain the operating margin for this division then overall company profitability should not be affected.

Corinna Berg, another analyst with the fund, reminds Petersen that during the current year, the US dollar weakened against the euro by 4% and that 50% of the sales in the recreational products division are sold in the United States.

Petersen recalls that some of the recent global expansion was aimed at establishing operations in Ireland because its statutory corporate tax rate is lower than the German rate of 29.8%. If Petersen assumes that other tax credits were the same in the current year as the prior year, he can analyze changes in Rhine's effective tax rate to determine whether the geographic mix of the company's profits has changed in the current year.

Petersen finally examines the company's liquidity ratios, which are shown in Exhibit 2. Even though the company's current and quick ratio have improved, his interpretation of the changes in the company's cash conversion cycle is that the company's liquidity position has deteriorated.

Exhibit 2 Rhine AG Selected Ratios

Ratio	Current Year	Prior Year	2 Years Prior
Current ratio	2.31	2.17	1.16
Quick ratio	1.06	0.89	0.53
Accounts receivable turnover	5.82	6.08	6.11
Inventory turnover	3.78	3.91	4.09
Accounts payable turnover	5.71	5.78	5.60
Cash conversion cycle	95 days	90 days	84 days

Worried that the balance sheet-based and cash flow-based accruals ratios (not shown) raise some concerns about the possible use of accruals to manage earnings, Petersen asks Berg for advice on what further type of analysis he should do as a follow-up on this issue.

Which of the following is the *most* appropriate use of Berg's reminder about the US dollar versus euro exchange rate in the current year? Peterson should use the information:

☒ when evaluating management's historical performance.
Correct Answer Feedback: Correct. Analysts should consider the foreign currency effect on sales growth for evaluating management's historical performance. Foreign currency fluctuations are out of management's control, so management should not be held accountable for the fluctuations when their performance is evaluated.

☐ to confirm that the division's organic growth was less than 11.2%.
Incorrect Answer Feedback: Incorrect. The unadjusted growth is 11.3%. If the currency devalued on 50% of the sales, then organic growth must have been greater than 11.3%.

☐ to determine the exchange gains or losses included in net income.
Incorrect Answer Feedback: Incorrect. Whether the FX gains/losses are included in net income depends on the type of foreign subsidiary. The calculation of the FX gains and losses considers more than just the devaluation on sales.

53 Multiple Choice 1 / 1 point

The *best* conclusion Petersen can make about the geographic mix of Rhine's profit in the current year is that compared with the prior year the mix is:

☐ more domestic.
Incorrect Answer Feedback: Incorrect. The effective tax rate went down, implying that more business is done in lower tax regimes

☐ about the same.
Incorrect Answer Feedback: Incorrect. The current year effective tax rate on earnings is lower than in the prior year (see table), implying that more profits were earned in a lower tax jurisdiction. The foreign operations are in lower tax regimes, so it is reasonable to conclude that more of the profits were earned internationally.

(€ millions)	Current Year	Prior Year
Earnings before taxes	136.6	170.0
Net earnings after tax	109.9	132.3
Income taxes	26.7	37.7
Effective tax rate (Income taxes/EBT)	19.6%	22.2%

☒ more international.
Correct Answer Feedback: Correct. The current year effective tax rate on earnings is lower than in the prior year (see table), implying that more profits were earned in a lower tax jurisdiction. The foreign operations are in lower tax regimes, so it is reasonable to conclude that more of the profits were earned internationally.

(€ millions)	Current Year	Prior Year
Earnings before taxes	136.6	170.0
Net earnings after tax	109.9	132.3
Income taxes	26.7	37.7
Effective tax rate (Income taxes/EBT)	19.6%	22.2%

Vignette

Sunjet Airlines Ltd. (Sunjet), a US-based "no frills" carrier, has the following existing non-domestic operations.

Nanuk Air Inc. (Nanuk) is a Canadian carrier that Sunjet purchased several years ago. It provides service to remote mining operations in the Canadian north. The company has been very profitable, and it recently financed a renewal of its fleet of seaplanes with CAD-denominated long-term debt.

Sunjet Mexico SA (SunMex) was established on 1 April 2016 to facilitate expansion of the Sunjet network to six Mexican destinations. The company purchased hangar assets in Mexico that were 100% financed with MXN-denominated loans guaranteed by Sunjet, and operations began shortly thereafter. Although most of SunMex's revenues are generated in the US vacation travel market, the company also serves domestic Mexican passengers. These MXN-denominated sales amounted to approximately 10% of SunMex's fiscal 2016–17 revenues, and they are expected to remain below 15% in the future.

Today is 30 April 2017, one month after Sunjet's fiscal year end, and the CEO, Mark Napier, is meeting with the CFO, Lisa Cameron, to discuss the international operations. Napier mentions that he will be meeting next week with the president of SunMex to discuss the first year of operations. Cameron pulls out the draft year-end results (Exhibit 1) and some exchange rate data (Exhibit 2). She notes that the SunMex president's bonus is tied to fixed asset turnover and net income targets. She reminds Napier that the bonus thresholds are evaluated based on the translated USD-denominated financial statements rather than the MXN-denominated ones, and she promises to send the translated version once the results are finalized.

Exhibit 1 Summarized Draft Financial Statements for SunMex, Fiscal 2016–17 (MXN millions)

Cash and accounts receivable	47
Fixed assets (net)	370
Total assets	417
Current liabilities	33
Long-term debt	224
Common shares	160
Retained earnings	0
Total liabilities and shareholders' equity	417
Sales	140
Depreciation expense	20
Other expenses	120
Tax expense	0
Net income	0

Exhibit 2 Selected Exchange Rates

	USD per MXN	CAD per USD
31 March 2017	0.0513	1.352
Average for fiscal 2016–17	0.0538	1.340
1 June 2016	0.0625	1.300
31 March 2016	0.0625	1.303
Average for fiscal 2015–16	0.0855	1.312

Nanuk, which prepares its translated statements using the current rate method, is next on the agenda. Cameron reports that annual translated USD-equivalent sales are up from 22.3 million last year to 23.7 million this year. Cameron reminds Napier that one of the performance metrics in the bonus calculation for Nanuk's president is Nanuk's sales growth determined in the local currency. Napier asks Cameron to calculate that figure for the 2016–17 fiscal year.

Napier also asks Cameron what effect Nanuk's translated statements will have on Sunjet's other comprehensive income for the current year.

Under which translation method for non-domestic operations will SunMex's fixed asset turnover *most likely* be higher?

- ☐ The temporal method

Incorrect Answer Feedback: Incorrect. Under the temporal method, the fixed asset turnover is lower.

- ☒ The current rate method

Correct Answer Feedback: Correct. Fixed asset turnover is higher under the current rate method, as shown in the following table. Although sales are translated at the average rate under both methods, the fixed assets are translated at the historical rate under the temporal method and at the year-end rate under the current rate method. Because the peso has weakened since the assets were purchased, the translated value is lower under the current rate method, resulting in a higher fixed asset turnover rate.

	MXN (millions)	Temporal Method		Current Rate Method	
		Exchange Rate (USD per MXN)	USD (millions)	Exchange Rate (USD per MXN)	USD (millions)
Sales	140	0.0538	7.532	0.0538	7.532
Fixed assets (net)	370	0.0625	23.125	0.0513	18.981
Fixed asset turnover (Sales/Fixed assets)	---	---	0.326	---	0.397

- ☐ There will be no difference.

Incorrect Answer Feedback: Incorrect. This answer incorrectly assumes that fixed assets are translated at the same rate under both the current rate and temporal methods.

55 Multiple Choice 0 / 1 point

On translation, SunMex's USD-denominated net income *most likely* includes:

- ☐ no re-measurement gains or losses.

Incorrect Answer Feedback: Incorrect. This answer assumes that the translational adjustment is shown on the balance sheet, as is the case for the current rate method.

- ☒ a re-measurement loss of \$1.792 million.

Incorrect Answer Feedback: Incorrect. This is the re-measurement loss that would be calculated under the current rate method.

Balance sheet	MXN (millions)	Exchange rate (MXN/USD)	USD (millions)
Net assets	(417 – 33 – 224) = 160	0.0513	8.208
Common shares	160	0.0625	10.000
Retained earnings*	0	---	0
Re-measurement loss (OCI)	---	---	1.792
Total equity	160	---	8.208

* AOCI = 10.0 – 8.208 = 1.792

- ☐ a re-measurement gain of \$2.526 million.

Correct Answer Feedback: Correct. SunMex will be translated using the temporal method because the competitive environment that mainly determines SunMex's sales prices is that of the United States. Under this approach, monetary assets and liabilities are translated at the year-end rate, and non-monetary items are translated at historical rates.

Because it is SunMex's first year of operations, the retained earnings will include only the translated net income for the year. The translated net income will include the re-measurement gain or loss in accordance with the temporal method. As shown in the following table, the retained earnings balance required to balance the balance sheet is USD2.352 million.

Balance Sheet	Item Type	MXN (millions)	Exchange Rate (MXN per USD)	USD (millions)
Cash and A/R	Monetary	47	0.0513 (year-end)	2.411
Fixed assets (net)	Non-monetary	370	0.0625 (historical)	23.125
Total assets	---	417	---	25.536
Current liabilities	Monetary	33	0.0513 (year-end)	1.693
Long term debt	Monetary	224	0.0513 (year-end)	11.491
Common shares	Non-monetary	160	0.0625 (historical)	10.000
Retained earnings*	Used to balance	0	---	X = 2.352
Total liabilities and equity	---	417	---	25.536

* Retained earnings: $25.536 - (1.693 + 11.491 + 10.000) = \2.352 million

Under the temporal method, the income statement is translated at the average rate, except for expenses linked to non-monetary items on the balance sheet. For SunMex, depreciation is translated at the historical rate, but the rest of the income statement is translated at the average.

Income Statement	MXN (millions)	Exchange Rate (MXN per USD)	USD (millions)	
Sales	140	0.0538 (average)	7.532	
Depreciation expense	20	0.0625 (historical)	1.250	
Other expenses	120	0.0538 (average)	6.456	
Income before re-measurement gain or loss	---	---	(0.174)	
Re-measurement gain**	---	---	Y = 2.526	See calculation
Net income required = retained earnings	0	---	2.352	From earlier balance sheet calculations

** The re-measurement (translation) gain or loss = $2.352 - (-0.174) = 2.526$ gain

56 Multiple Choice 0 / 1 point

In the bonus calculation for Nanuk's president, the sales growth that is to be used is *closest* to:

☒ 6.3%.



Incorrect Answer Feedback: Incorrect. This is the USD-denominated growth rate of 6.3% (USD1.4 million/USD22.3 million).

☐ 8.5%.



Correct Answer Feedback: Correct. Nanuk's sales growth, in the local Canadian currency, is its CAD-denominated sales growth.

	USD (millions)	Average Exchange Rate (CAD per USD)	CAD (millions)
2016–17 revenues	23.7	1.340	31.76
2015–16 revenues	22.3	1.312	29.26
Increase in sales	1.4	---	2.5

CAD-denominated sales growth is $2.5 \text{ million} / 29.26 \text{ million} = 8.5\%$.

☐ 10.3%.

Incorrect Answer Feedback: Incorrect. This is the CAD-denominated growth rate incorrectly translated back to CAD using year-end exchange rates: $\text{CAD}2.98 \text{ million} / \text{CAD}29.06 \text{ million} = 10.3\%$.

	USD (millions)	Year-end exchange rate (CAD per USD)	CAD (millions)
2016/17 revenues	23.7	1.352	32.04
2015/16 revenues	22.3	1.303	29.06
Revenue growth	1.4	---	2.98

57 Multiple Choice 1 / 1 point

The *best* answer to Napier's question about the effect of Nanuk on Sunjet's other comprehensive income is that Nanuk's:

☒ net asset exposure will generate a re-measurement loss.



Correct Answer Feedback: Correct. Nanuk is translated under the current rate method, so its translational exposure is its net asset position. The weakening CAD (see Exhibit 2) will generate a re-measurement loss in Sunjet's other comprehensive income.

☐ net asset exposure will generate a re-measurement gain.

Incorrect Answer Feedback: Incorrect. It is the net asset position that is exposed to exchange fluctuations under the current rate method. Per Exhibit 2, the CAD is depreciating vs. the USD so would generate a loss. Candidates may think the CAD is strengthening.

☐ net liability exposure will generate a re-measurement gain.

Incorrect Answer Feedback: Incorrect. Nanuk's net monetary liability position would generate a re-measurement gain under the temporal method, but it is the net asset position that is exposed to exchange fluctuations under the current rate method.

Vignette

Marcus Eriksson, chief financial officer of Trana AB, and Katrina Lars, director of financial reporting, are preparing the company's 2015 annual report. Today's meeting is to discuss the transactions and disclosures related to Trana's foreign operations. Trana, which reports under International Financial

Reporting Standards (IFRS), is a Sweden-based retailer operating stores in three geographic locations: Sweden, the eurozone (with a current presence only in France, Germany, and Italy), and the United States. The stores in the eurozone and the United States are operated through a wholly owned subsidiary in each region. Consistent with Swedish accounting practice, the annual report includes separate financial statements for the parent company (Trana) and consolidated, or group, financial statements. The income statements are presented in Exhibit 1.

Exhibit 1 Trana AB Income Statements for the Years Ended 31 December

	Consolidated Income Statement (SEK millions)		Parent Income Statement (SEK million)	
	2015	2014	2015	2014
Sales	30,200	26,892	4,700	4,653
Cost of goods sold	13,590	12,639	2,600	2,650
Gross profit	16,610	14,253	2,100	2,003
Selling expenses	10,872	9,143	1,500	1,525
Admin. expenses	1,510	1,076	260	200
Operating profit	4,228	4,034	340	278
Net other costs/losses	10	15	2	3
Earnings before taxes	4,218	4,019	338	275
Income taxes	1,122	1,071	74	61
Net profit	3,096	2,948	264	214

Eriksson and Lars start the meeting by reviewing some of the relevant currency exchange rates, shown in Exhibit 2. The functional currency for the eurozone and US subsidiaries is the local currency (EUR and USD, respectively), thus the financial statements of both are translated using the current rate method. Both subsidiaries are consistently profitable.

Exhibit 2 Exchange Rates

	SEK per EUR	SEK per USD
Beginning 2014	8.43	6.32
Average 2014	8.555	6.35
End of 2014	8.88	6.38
Average 2015	9.125	7.595
End of 2015	9.31	8.81

Next, they review the performance and related disclosures by region. The number of stores operated in each region is shown in Exhibit 3.

Exhibit 3 Number of Stores by Region

Year	Eurozone	Sweden	United States	Total
2014	340	99	80	519
2015	400	100	80	580

In preparation for the meeting, Lars looked at the US region and calculated the effect of the change in the SEK/USD exchange rate on the increase in sales from 2014 to 2015. Her notes include the following:

- In 2014, the sales per store, in SEK, were the same for both US and Swedish stores.
- The sales per US store in USD remained constant in 2015.

Eriksson reminds Lars that Trana defines organic growth in retail as coming from two factors:

1. increasing the number of stores, and
2. increasing the sales per store in the local currency.

He says that he wants to provide disclosures related to the organic growth rate in domestic sales per store, by region, and asks Lars to calculate it for the eurozone region where the sales figures (in millions) were SEK18,394 in 2014 and SEK21,640 in 2015.

In 2012, at the start of Trana's expansion into North American markets, the company established a subsidiary, Anart Inc., in a South American country to benefit from lower labor and shipping costs. The details of the Anart investment are as follows:

- Anart is 80% owned by Trana with 20% local investment.
- It sells all of its production to Trana and Trana's other subsidiaries and determines the transfer price as full cost plus 5%.
- In 2015, sales (in millions) from Anart to Trana companies were SEK4,485 with net profit of SEK204.
- The corporate tax rate in the country is 10%.

Throughout 2013, the South American country experienced high rates of inflation, approaching 30% per year. Trana had originally assumed that the high inflation rate was temporary, but it has shown no signs of decreasing and is now a concern. Eriksson and Lars discuss the impact of Anart on Trana's financial statements and Eriksson asks Lars:

"Is the same accounting method being used this year to account for Anart in the consolidated financial statements as in prior years?"

Eriksson reminds Lars that there is a proposal in Sweden to reduce the corporate tax rate from the current 22% to 16.5%. He would like to provide pro-forma disclosures related to the potential change in net income this change could provide for Trana. He reminds Lars that the average tax rate for the eurozone countries where Trana operates is 30% and 25% in the United States. Sweden operates under a tax treaty with all countries in which it has subsidiaries, such that it will owe taxes on foreign earned income to the extent that the Swedish rate exceeds the foreign rate.

In closing the meeting, Eriksson mentions that Trana is undertaking a comprehensive review of its operations in 2016, and its objectives include reducing overall tax costs by lowering its effective tax rate and reducing foreign exchange gains and losses reported on the income statement.

58 Multiple Choice 0 / 1 point

Using Exhibits 1, 2, and 3 and Lars's notes about the US operations, the change in sales reported for the US region (in SEK millions) explained by the change in the SEK/USD exchange rate in 2015 is *closest* to:

☐ SEK737.

Correct Answer Feedback: Correct. The number of stores in the United States is the same in 2014 and 2015 (80). The average sales per US store in 2014 is the same as the Swedish stores, and the USD sales are the same in both years. But when sales are converted into SEK, the values reflect the change in the exchange rate over the period.

	Calculations	SEK (millions)
Number of stores in the United States (Exhibit 1)	80	---
Average sales/US store in 2014 (same as the Swedish stores in 2014)	SEK4,653/99 stores =	47/Store
Sales/US store (in USD millions) based on average exchange rate in 2014	(SEK47/Store)/(SEK6.350/USD) = USD7.40/Store	---
Total USD sales (in USD millions) in 2014 and 2015 (same both years in USD)	80 stores × (USD7.40/Store) = USD592.13	---
Total USD sales in 2014 in SEK	USD592.13 × (SEK6.350/USD) =	3,760
Sales in USD in 2015 (the same as 2014), converted at the 2015 average rate	USD592.13 × (SEK7.595/USD) =	4,497
Increase in sales because of change in exchange rate		737

☐ SEK813.

✗

Incorrect Answer Feedback: Incorrect. It uses the average sales overall, not the Swedish ones to determine sales level.

Number of stores in the United States	80	---
Average sales/US store (in millions SEK) is the same as total stores in 2014	In 2014: SEK26,892/519 store = SEK51.815/store	(the error)
Sales/US store (in millions USD) based on average exchange rate in 2014	SEK51.815/store/6.35 SEK/USD = \$8.160/store	---
Total USD sales (in millions USD) in 2014 and 2015 (same both years in USD)	80 store × 8.160 USD/store = 652.79 USD	---
Total USD sales in 2014 (in millions SEK)	652.79 USD × 6.350 SEK/USD	SEK4,145
Sales in USD in 2015 (the same as 2014) Converted at the 2015 average rate	652.79 USD × 7.595 SEK/USD	SEK4,958
Increase in sales (in millions) due to change in exchange rate		SEK813

☐ SEK1,432.

Incorrect Answer Feedback: Incorrect. It uses year-end rates instead of average rates to convert sales

Number of stores in United States	80	---
Average sales/US store (in millions SEK) (same as the Swedish stores in 2014)	In 2014: SEK4,653/99 stores = SEK47/store	---
Sales/US store (in millions USD) based on ending exchange rate in 2014	SEK47/store/6.380 SEK/USD = \$7.367	(the error)
Total USD sales (in millions USD) in 2014	80 stores × \$7.367/store = \$589.36	---
Total USD sales in 2014 (in millions SEK)	\$589.36 × 6.380 SEK/USD	SEK3,760
Sales in USD in 2015 (the same as 2014) Converted at the 2015 closing rate (the error)	\$589.36 USD × 8.810 SEK/USD	SEK5,192
Increase in sales (in millions) due to change in exchange rate		SEK1,432

59 Multiple Choice 1 / 1 point

Using Eriksson's definition, the organic growth rate in sales per store in the eurozone region between 2014 and 2015 that Lars calculates is *closest* to:

☒ -6.2%.

✓

Correct Answer Feedback: Correct. To reflect the growth in domestic sales per store, it is necessary to eliminate the foreign exchange effect.

	2014	2015
European sales (given) (in millions)	SEK18,394	SEK21,640
Exchange rate (average rate used for sales under current method)	SEK8.555/EUR	SEK9.125/EUR
Sales	EUR2,150	EUR2,372
Number of stores	340	400
Sales/Store	EUR6.32/Store	EUR5.93/Store
Growth rate $(5.93 - 6.32)/6.32$		-6.2%

☐ 0%.

Incorrect Answer Feedback: Incorrect. This does not adjust for the FX.

	2014	2015
European sales (in millions)	SEK18,394	SEK21,640
# of stores	340	400
Sales/store in SEK	SEK54.10	SEK54.10
Growth rate (no change)		0.0%

☐ 10.3%

Incorrect Answer Feedback: Incorrect. It is the growth rate in Euros in total, not adjusting for the increase in the number of stores.

$(2,372 - 2,150)/2,150 = 10.3\%$:

	2014	2015
European sales (in millions)	SEK18,394	SEK21,640
# of stores	340	400
Sales/store in SEK	SEK54.10	SEK54.10
Growth rate (no change)		0.0%

60 Multiple Choice 1 / 1 point

Which of the following is Lars's *most* appropriate answer to Eriksson's question concerning the accounting method used for Anart in 2015?

☐ Yes, the temporal method is being used, as in past years.

Incorrect Answer Feedback: Incorrect. Now that the high inflation has lasted at least three years it can be considered hyperinflation, and different translation methods must be used to reflect economic reality.

☒ No, the current rate method is being used, after restating nonmonetary items for inflation.

Correct Answer Feedback: Correct. Because Anart is an extension of Trana (Anart sells 100% of its production to the group) its functional currency would be the Swedish krona, not the local currency, and it would be considered an integrated foreign operation. As an integrated foreign operation, Trana would normally, and historically, have accounted for Anart using the temporal method. But the country in which Anart operates is experiencing high inflation; three years (2013–2015) of rates near 30% would exceed the 100% indicator of hyperinflation. Therefore, under IFRS, the nonmonetary items must be adjusted for the loss in purchasing power to better reflect economic reality. Note that only the nonmonetary items are adjusted because monetary ones would already be expressed in the monetary unit current at the balance sheet date.

☐ No, the current rate method is being used, after restating all accounts for the general price index.

Incorrect Answer Feedback: Incorrect. Only nonmonetary items are affected by the loss of purchasing power and must be restated.

61 Multiple Choice 0 / 1 point

If the proposed reduction in Swedish tax rates had been in effect in 2015, the increase in Trana's net profit (in SEK millions) would have been *closest* to:

☐ SEK18.6.

Incorrect Answer Feedback: Incorrect. It is only the EBT in Sweden: $338 \times (0.22 - 0.165) = 18.6$.

☒ SEK29.8.

Incorrect Answer Feedback: Incorrect. It uses net profit as given for South America and fails to adjust it back to EBT: $(338 + 204) \times (0.22 - 0.165) = 29.81$.

☐ SEK31.2.

Correct Answer Feedback: Correct. The proposed change in Swedish tax rates would have affected the income earned in Sweden (SEK338 before tax) and the pre-tax income earned in the South American subsidiary (SEK227, see calculation in following table) because the tax rate there is lower than in Sweden and hence subject to tax at Swedish rates. The income earned in tax jurisdictions with rates higher than Sweden's (Europe and the United States) are not subject to tax in Sweden and thus would not have been affected.

		Tax Effect as Reported (22%)	Under Proposal (16.5%)	Difference
Swedish earnings before taxes (EBT)	SEK338	74.4	55.8	SEK18.6
South American EBT*	SEK227	50.0	37.4	SEK12.6
Total	---	---	---	SEK31.2

* To calculate EBT, divide net profit of SEK204 by $(1 - \text{tax rate})$: $204 / (1 - 0.10) = \text{SEK}227$.

62 Multiple Choice 0 / 1 point

Which of the following strategies would be *most likely* to help Trana achieve at least one of the objectives mentioned by Eriksson for 2016?

☐ Increase the number of stores in the US region

Incorrect Answer Feedback: Incorrect. Increasing the number of stores in the US region will not affect the tax rate, but would increase taxable income because the tax rate there is greater than in Sweden and would not affect foreign exchange gains and losses on the income statement because it is self-sustaining, and the gains and losses go to other comprehensive income.

☒ Initiate a hedge on the net asset position of the eurozone subsidiary

Incorrect Answer Feedback: Incorrect. The eurozone subsidiary is also self-sustaining, and any effect of hedging its net asset position would go to other comprehensive income, not net income. The eurozone's taxes are higher than in Sweden, so there would be no lowering of taxes either.

☐ Raise the price at which Anart sells its goods to other group members



Correct Answer Feedback: Correct. Anart operates in a South American country with the lowest tax rate of the group—10% versus 25% in the United States, 30% in the eurozone, and 22% (or 16.5%) in Sweden. If more of the corporate profits are earned by Anart, the effective tax rate will decrease.

- Anart currently earns a return of $204/4,485 = 4.5\%$, whereas the overall corporate profit rate is 10.3% ($3,096/30,200$).
- Any income taxed in South America would be eligible for a tax credit in Sweden, and Trana would be liable for the tax difference between the local 10% rate and the rate in Sweden (22% or 16.5%).
- To the extent that taxable income can be diverted from the US or eurozone operations (where the rates are higher than Sweden's), it would result in an overall tax saving for Trana.
- By increasing the price at which Anart sells goods to the US and eurozone subsidiaries, it would increase the taxable income earned in South America and reduce the taxable income (through higher cost of goods sold) in the United States and the eurozone. Because of the tax treaty with Sweden, there would be no net tax savings on the goods sold to US and eurozone stores by Anart if the prices change.

Because both retail subsidiaries are translated using the current rate method, all foreign exchange gains/losses are reported in other comprehensive income not on the income statement. Therefore, the effects of hedging the exposure in the eurozone subsidiary would also be reported in other comprehensive income and not affect the income statement.

Increasing the number of stores in the US would increase the amount of income in the highest tax jurisdiction and hence increase taxes, not lower them.